Arnav Rajashekara

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EDUCATION

Cornell University, Ithaca, NY. **– M.Eng.** Computer Science. **Graduation:** May 2027
The Ohio State University, Columbus, OH. **Graduation:** May 2026

- **B.S.** Computer Science & Engineering (**AI Track**), Minor in Statistics. **Cum Laude with Honors Research Distinction**.

WORK EXPERIENCE

AI Recruiting Stealth Startup (*Founding Software Engineer Intern)* **May 2026 – Present**

- Reduced data collection latency by approximately 90% by architecting a data pipeline covering multiple major platforms, with concurrent processing to support large-scale, ongoing collection.
- Engineered a resilient scraping system combining API-based and browser-based collection methods (Scrapy, Playwright) to reliably gather data from both static and JavaScript-heavy sites.
- Designed a dual-cadence scheduling architecture combining GitHub Actions cron (daily) with a persistent watchdog container (hourly hash-based change detection), backed by Uptime Kuma monitoring for real-time failure visibility.

American Institutions & Methodology Lab – OSU (*Research Assistant – Team Lead)* **Aug 2024 – May 2026**

- Led a research team to develop an issue classification model that categorizes SCOTUS petitions into issue areas.
- Applied natural language processing techniques and unsupervised machine learning algorithms, including Legal-BERT embedding generation, tokenization, and K-Means clustering, to uncover latent structures in legal text.
- Selected as 1 of 15 total recipients of the Undergraduate Research Apprentice Program (URAP) and awarded a total of nearly $6,000 to support the project. Project awarded over $500,000 in grants by the National Science Foundation (NSF).
- Earned highly competitive Honors Research Distinction award. Granted a scholarship of $2200 and wrote and defended an undergraduate thesis on original research.

American Honda Motor Company (*Systems & Analytics Engineer Intern)* **Aug 2025 – Dec 2025**

- Projected to save up to $100,000 annually by delivering KPI dashboards leveraging live production data to enhance process monitoring and bottleneck detection.
- Prototyped and engineered industrial production applications using Figma for UI/UX design and Ignition SCADA for real-time process visualization, integrating user feedback through iterative testing cycles.
- Selected as a finalist for "Intern of the Term" out of 130+ interns. Awarded to interns for exceeding expectations and driving meaningful improvements to team deliverables.

NetJets – Berkshire Hathaway Private Aviation Company (*Software Engineer Intern)* **May 2025 – Aug 2025**

- Developed a secure, scalable, and intuitive web-based portal for business analysts to view AWS S3 data.
- Built the client-side interface using ReactJS, supported by AWS Amplify for backend services and deployment.
- Leveraged Terraform to define and manage the cloud infrastructure, enabling modular deployment and scalability.
- Increased data security by leveraging AWS IAM Identity Center in conjunction with RBAC and MFA via Okta SSO.

Mimecast – AI Powered Risk Management *(Software Engineer Intern)* **May 2024 – Aug 2024**

- Decreased page load time by 67% by implementing client-side caching with React Query for tenant profile data.
- Built a new customer profile configuration workflow using ReactJS, Next.js, and Tailwind CSS.
- Utilized Java and Cucumber to implement back-end solutions and develop testing frameworks to ensure full functionality.

PROJECTS

ApexClip *(Solo Project)* **May 2026 - Present**

- Engineering an automated video analysis pipeline using OpenCV and YOLOv8 object detection to capture exciting moments within sports video and turn it into a highlight video.
- Designing a multi-threaded temporal buffer layer to store video frames in memory, allowing the engine to slice seamless video segments around high-motion triggers.
- Formulating frame-by-frame filtering algorithms and confidence thresholding to optimize processing throughput and isolate action segments from raw footage.

StockRag *(Team of Six Engineers)* **Jan 2026 – April 2026**

- Architected a Retrieval-Augmented Generation (RAG) pipeline utilizing a Python (FastAPI) backend and a TypeScript React frontend to deliver real-time financial insights.
- Engineered an automated data extraction and parsing workflow to ingest market metrics, indexing tokenized embeddings into a Chroma vector database for high-fidelity document retrieval.
- Developed async hooks and state orchestration layers (useChat.ts) to manage low-latency streaming chat interactions, linking frontend telemetry to the core LLM orchestration service.

TECHNICAL SKILLS

- **Languages & Cloud:** Python, TypeScript, Java, SQL, C, R, MATLAB, HTML/CSS, AWS, Azure, Terraform, Docker
- **Frameworks & Libraries:** FastAPI, React, Next.js, Node.js, Tailwind CSS, Shadcn UI, Radix UI, Pytest, Cucumber
- **AI & Data:** PyTorch, TensorFlow, YOLOv8, OpenCV, Chroma Vector DB, Scrapy, Playwright, Pandas, LangChain